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Project Description

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Table 2: Sample used for the simple dynamic empirical model.

Statistic	Value
Households	42,816
Weeks	52
Household-weeks	2,226,432
Observed purchases	301,027
Purchase rate	13.5%
Brand 1 purchases	66,446
Brand 2 purchases	234,581
Duration cap	12
Discount factor	0.950

	Unnamed: 0	Customer_ID	week_num	y	Brand	Quantity	promo_flag	Price	last_purchas
0	1	1	1	0	0	0	0	NaN	NaN
1	2	1	2	0	0	0	0	NaN	NaN
2	3	1	3	0	0	0	0	NaN	NaN
3	4	1	4	0	0	0	0	NaN	NaN
4	5	1	5	0	0	0	0	NaN	NaN

Simple Dynamic Empirical Model

The original CML estimator is quite demanding because it only uses very specific purchase histories with matching future price states. As a simpler empirical alternative, we estimate a finite-horizon dynamic logit model directly on the scanner panel. The model keeps all weeks 1–52, but compresses the state space by capping the number of weeks since the household's last purchase at 12.

The household chooses among no purchase, brand 1, and brand 2. The state is the current week, the household's last purchased brand, and the number of weeks since that purchase. Prices and promotions are treated as observed weekly state variables. The dynamic component is estimated by nested fixed point: for each parameter vector, the value function is solved by backward induction and the resulting conditional choice probabilities are used in the likelihood.

Table 3: Simple empirical dynamic-logit estimates.

Model	Parameter	Estimate	Std. err.	z-stat	p-value	CI 2.5%	CI 97.5%
Dynamic	No-purchase constant	-0.833	0.008	-102.321	0.000	-0.849	-0.817
Dynamic	Brand 2 constant	-0.248	0.004	-60.087	0.000	-0.256	-0.240
Dynamic	Price coefficient	-0.224	0.001	-361.977	0.000	-0.225	-0.223
Dynamic	Switching cost	0.953	0.005	183.213	0.000	0.943	0.963
Dynamic	Duration penalty	0.004	0.000	32.137	0.000	0.004	0.004
Myopic	No-purchase constant	-0.726	0.008	-89.807	0.000	-0.742	-0.711
Myopic	Brand 2 constant	0.001	0.005	0.144	0.885	-0.009	0.011
Myopic	Price coefficient	-0.231	0.001	-367.108	0.000	-0.233	-0.230
Myopic	Switching cost	0.810	0.005	152.910	0.000	0.800	0.820
Myopic	Duration penalty	0.000	0.000	10.057	0.000	0.000	0.000

Table 4: Fit comparison for the simple empirical model.

Model	Converged	Log-likelihood	AIC	BIC	Iterations
Dynamic	Yes	-926410.153	1852830.305	1852893.385	35
Myopic	Yes	-929954.783	1859919.566	1859982.646	39